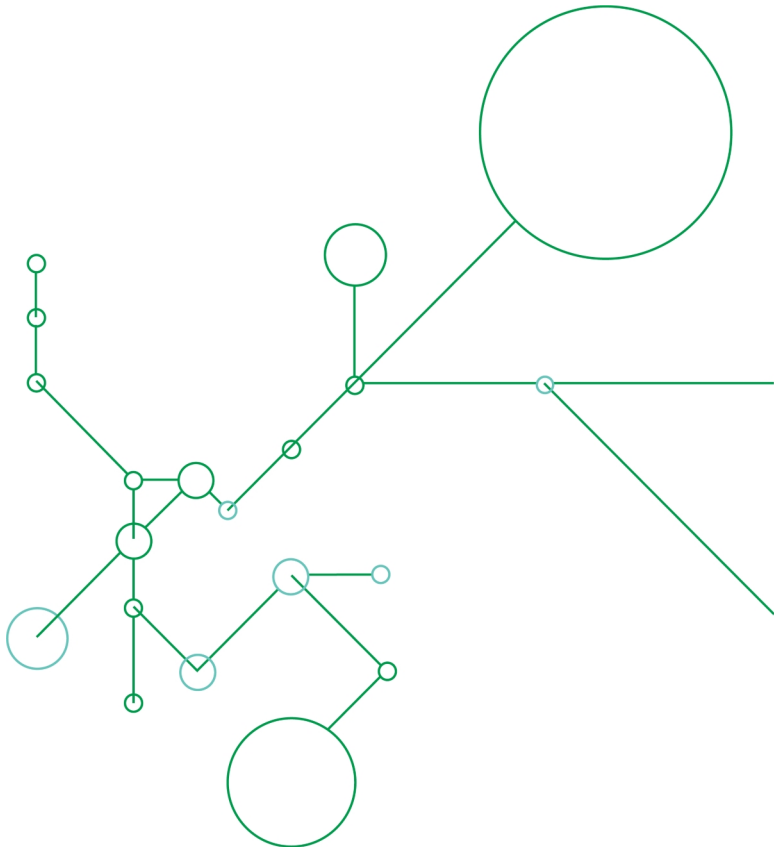
Business Cases Book

Create Visualizations with Qlik Sense®



Qlik Sense

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Case 1: Analyzing business data

Duration: (~1 hour)

Summary:

- Practice navigation and selections in a Qlik sense app.
- Perform searches and analyze the visualizations to answer business questions.

Modules:

- Overview, selecting and searching data

Task description

As a business analyst, you need to be able to analyze a Qlik Sense app by making selections and searching in the data.

These industry-specific apps provided by the business need some answers to the business questions related to them.

Healthcare: Clinical Variation - Orthopaedics

App link: https://demos.qlik.com/qliksense/ClinicalVariationOrthopaedic

Questions:

1. Analyze the mortality rate for female patients in emergency procedures across all the hospitals. Which hospital has the highest rate? Which hospital has the lowest?
2. How many clinicians work in several hospitals? Could you name the hospitals and the clinicians who work in more than one hospital?
3. What are the top 3 procedures with the longest Average Length of patient Stay (ALOS)? How many clinicians apply these procedures?
4. How many episodes are registered for patients between 31 and 36 years old, with severe or catastrophic complexity? What is the ALOS and the complication rates for those patients?
5. What is the patient ID with the highest number of procedures? How many times has this patient been discharged? Could you provide more details about the patient (gender, age) and the injuries (diagnosis details, clinician, hospital)?

Human Resources: Human Capital Management

App link: https://demos.qlik.com/qliksense/HumanCapitalManagement

Questions:

1. What is the role with the highest number of employees in the company? How many employees are in this role and what's their satisfaction ratio? How many of them are active?
2. Based on the employees' tenures, how many employees with the longest tenure are in the company and where are they based (office location)? Are they all full time (FTE) or are there any part-time employees (PTE) among those?
3. The company has invested $767k in employee development and training programs. In the United Kingdom, analyze if the programs are being completed (having more than a 50% training completion). Which departments need to focus on improving employee participation?

4. Regarding salaries, which employee band has the highest wages? Are women and men paid similarly in this band? In which countries does a woman get paid (on average) higher wages than a man?
5. Which employee age range has salaries higher than 124k? Where are those employees based and what's the distribution across gender and ethnicity? Review the Diversity workforce analysis story. What else can you tell about the diversity in the company?

Sales: Executive Dashboard

App link: https://demos.qlik.com/qliksense/ExecutiveDashboard

Questions:

1. In order to reduce travel expenses, review the percentage of travel expenses that are foreign and the percentage that are local. Are these percentages different from the previous year? Are those expenses above or below the target?
2. Which product category produces more revenue in Americas for the supermarket segment? In that product subgroup, identify the top products with higher revenues in the region for that segment. Are there any products in that category generating negative margin%?
3. The company would like to analyze the expenses in the last year versus the previous year. How have the different regions done in cutting expenses? Apart from general costs and staff costs, what other costs do the companies have and what percentage of sales do they represent (analyze these per each year)?
4. What is the total amount overdue for convenience stores? Which customers with overdue amounts, have an average sales per day higher than 1k? What is the outstanding average sales per day for those customers?
5. Analyze the trend of sales per day for alcoholic beverages during the period of December 2016 to March 2017. What does this trend look like for the Hotels & Clubs and Restaurant & Cafes segments? Which sales rep generated the highest sales for these segments within this date period closer to the margin target? Where is this sales rep based (region)?

Retail: Supply Chain - Operations, Orders, and Logistics

App link: https://demos.qlik.com/qliksense/SupplyChainOperations

Questions:

1. The company wants to analyze the ground vs. air shipments for customers based in the United States. What are the percentages of ground and air orders made through the online sales channel for those customers? For this same scenario, what is the percentage of on-time shipments on these two delivery methods?
2. How many shipments originated by the web channel (company primary channel)? How many shipments are made in that channel by customers between 35 and 42?
3. Customer satisfaction is one of the priorities of the company. Is there any product group that is an outlier with respect to on-time percentage and ages? Analyze the orders for that product and review the shipment method. Which order and shipment method was used for those orders? Where is the customer based?
4. What other product groups were bought by companies who bought the LP grills? Which companies are buying gas trimmers? What other products are they buying? What tier are these customers?
5. The manufacturer of the "HE ELECTRIC DRYER" has informed the company that the product is faulty and they need to send a message to all the customers who bought this product. The

current open orders for this product will be canceled. Which customers were expecting this product? Find out the customers' details (contact preference, customer name, city, and country).

IT: Help Desk Management

App link: https://demos.qlik.com/qliksense/HelpdeskManagement

Questions:

1. How many high priority cases are over a month old? Identify the case owners with high priority open cases, with an average case duration of 30 or more days. In what status are these cases?
2. How many open cases with high priority are duplicate cases?
3. The company launched the new intranet in 2016 Q1. When did the web infrastructure department reach the highest number of cumulative new cases in that period?
4. The systems department would like to review all the high and medium priority cases opened between 15th March 2015 and 15th April 2015, in the web infrastructure department. Based on the highest case duration, analyze the subjects of the top 3 cases. Are there any other cases in that period with those subjects, but with less duration?
5. The company would like to analyze all the high and medium priority cases that have been open for greater than a month. They would like to understand who is responsible for those cases (who has the most opened cases with these criteria) and the status of those cases.

Process description

1. Choose one of the apps to analyze. Then, open the app and navigate through the sheets.
2. Get familiar with the app data. Review the visualizations, dimensions, measures, and read all the description information.
3. Read the business questions and, using the app, make selections and search to answer those.

Apps and data files

- All the apps are available in the Qlik Sense demo site (demos.qlik.com/qliksense). In case you cannot access the demo site, the apps are also available for download under **Student Files**.

  Qlik Sense Client-Managed: If you are working with the Skytap environment, the app or apps will be available under the **Everyone** stream.

- No data files are needed for this business case.

Hints

- Review all the sheets, visualizations, stories, and bookmarks (if available) in the app.
- Make selections in visualizations. Use the visual search and smart search to find the areas in the data you need to focus on.
- Individual fields' values can also be selected using the selections panel.

Case 2: Loading business data

Duration: (~1.5 hours)

Summary:

- Add and transform data in a Qlik Sense app using the Data manager.
- Associate tables and verify the data associations using selections.
- Create master items and visualizations to answer business needs.

Topics:

- Preparing data for analysis
- Creating visual analytics

Task description

You are a business analyst working for a call center company. The company has registered all the calls during the last period and would like you to analyze the following KPIs:

- Number of calls
- Number of customers
- Average call satisfaction (as a percentage)
- Average call duration (in seconds)

They also want to see those indicators based on the sector, customer age group, operator, call center location (city and country), and specific periods (years, quarters, and year months).

An Excel file has been provided with all the business data: **CallCenterDataset.xlsx**.

Create a Call Center Dashboard using Qlik Sense and the data provided to answer the following business questions:

- What does the trend over time (by year, month, and day) look like for the number of calls and the average percentage of call satisfaction?
- Who are the top 10 operators, based on the number of calls? And based on call duration? Also highlight the average call satisfaction for them.
- What is the average percentage of satisfaction geographically (based on the call center country)?
- What is the composition of customers based on sector? What is the composition of customers based on the following age ranges: less than 30, 30-40, 41-60, more than 60?
- Is there a correlation between the percentage of satisfaction and the call duration per sector? Also show the number of customers.

The company has provided metrics to validate the numbers when loading the data. Make sure the figures match your results:

Country	Calls	Customers	Avg Call Duration	Avg Call Satisfaction
Totals	**2000**	**570**	**778.4875**	**49.57%**
United States	445	296	779.7236	49.75%
United Kingdom	248	196	808.4032	52.35%
Argentina	239	191	758.0460	47.57%
Sweden	236	201	819.1610	54.76%
Portugal	224	185	803.5938	50.05%
Germany	206	179	798.6699	44.55%
Switzerland	202	170	715.6683	48.94%
Italy	200	165	729.6150	47.23%

Sector	Calls	Customers	Avg Call Duration	Avg Call Satisfaction
Totals	**2000**	**570**	**778.4875**	**49.57%**
Other	576	166	789.3646	49.18%
Services	305	78	803.6492	50.58%
Retail	297	84	742.1717	48.66%
Government	286	83	803.6154	47.86%
Healthcare	270	79	778.7037	51.08%
Cross-Industry	266	80	739.3947	50.56%

OpName	Calls	Customers	Avg Call Duration	Avg Call Satisfaction
Lucas Goldberg	128	113	742.5000	55.15%
Luke Smith	127	112	820.8819	48.62%
Sarah Turner	114	107	827.1491	53.35%
Robert Anderson	112	102	733.0804	54.51%
Jennifer Soto	110	99	825.4000	50.45%
Alex Byrd	110	99	758.4818	42.73%
Adam Jones	109	101	792.1101	51.32%
Claire Cooke	108	100	842.6111	45.57%
Ada Persson	107	99	842.8505	48.80%
Joe Kent	102	97	774.3725	53.96%
Kevin Smith	102	96	775.9118	48.15%
Sam Walker	101	92	720.3366	57.43%
Rose Bain	100	96	793.8500	43.74%
Joe Williams	100	91	822.0600	49.14%
Oscar Bell	97	91	713.1031	48.91%
Peter Grant	96	89	763.4792	48.49%
Emily Miller	95	84	710.9789	49.31%
Victor Gray	92	84	789.7717	47.68%
Sophie Milson	90	82	715.4222	41.68%

Process description

1. Review the data provided in the Excel file and list the data transformations you will need to create. Note, there might be data quality issues.
2. Create a new Qlik Sense app and load the data.
3. Transform the data and associate the tables so the associations in the final dashboard work as expected.
4. Create master measures and master dimensions based on the business needs.
5. Create one sheet for the call center dashboard and use the most effective visualization in each case based on the business questions.

Apps and data files

- No apps are needed for this case. You will create a new app during the business case.
- There is only one data file called **CallCenterDataset.xlsx**. You can download the file from the course page: **Student Files > Case2-LoadingBusinessData**.

> **Qlik Sense Client-Managed**: If you are working with the Skytap environment, this file already exists under **C:\Modules_Create Visualization with Qlik Sense\CVQS_BusinessCases\Case2-LoadingBusinessData**.

The file contains three tables:

- **Calls** (contains the list of calls, with all the details related):

Field	Description (type or format)
CallID	Unique call identifier (integer)
CallStarts	Date and time when the call started (*DD/MM/YYYY hh:mm:ss*)
CallEnds	Date and time when the call ended (*DD/MM/YYYY hh:mm:ss*)
CustomerAcc	Customer account identifier (alphanum, *ACCXXXX*)
CustomerAge	Customer age (number, integer)
CustomerPhone	Customer phone number including prefix (*+XX XX-XXX-XXX*)
OperatorID	Operator identifier who made the call (alphanum, *P00XX*) Links with Person table.
LocationID	Location identifier for the call center location (alphanum, *LOCXXX*) Links with Location table.
Sector	Area or industry related to the call: *Retail*, *Healthcare*, etc. (string)
CallSatisfaction	Satisfaction score the customer provided after the call (decimal)

- **Location** (list of call center locations): includes the unique identifier, name, city, and country of the call center.
- **Person** (list of the operators making the calls): includes the unique identifier, first name, and surname of the operator.

Other definitions the business provided are:

- **Number of calls** is the count of *CallID* values.
- **Number of customers** is the count of distinct *CustomerAcc* values.
- **Call duration** is the time (hh:mm:ss) between the call start and end: *Time(CallEnds - CallStarts,'hh:mm:ss')*.

  The call duration needs to be displayed in seconds: *Round(CallDuration * 86400)*
- **Percentage of call satisfaction** is the *CallSatisfaction* multiplied by 100.
- **Call time periods** are the year, quarter, month, or date based on the *CallStarts* timestamps.
- **Customer age range** is the *CustomerAge* values split into buckets:

Customer Age	Customer age range
Age < 30	Less than 30
Age between 30-40	30-40
Age between 41-60	41-60
Age > 60	More than 60

Hints

- Use Data manager to transform and associate the data:
  - Create new calculated fields needed in the dashboard (i.e., *CallDuration* and *OperatorName*).
  - Fix any data quality issues (i.e., *Country* and *LocationID* fields).
  - Create field buckets (i.e., *CustomerAge* field).
  - Make sure the date fields and geoinfo fields are automatically created.
- Create master measures using the following aggregation functions: **Count()** and **Avg()**.
- Use the following visualizations: KPI, Map, Combo chart, Bar chart, Scatter plot, and Treemap. Also, include filter panes with the most relevant filters.
- You can create alternative dimensions or measures and use drill-down groups if needed. For instance, you can show the trend based on years, then drill down to quarter or month, then to day.
- When needed, use color to code the values in visualizations by measure or by dimension.

Case 3: Advanced business analysis

Duration: (~3.5 hours)

Summary:

- Add and transform data in a Qlik Sense app using the Data manager.
- Associate and concatenate tables.
- Include additional data to cover the business needs.
- Use visualizations with advanced expressions to answer business questions.
- Create a story to reveal your analysis using snapshots and bookmarks.

Topics:

- Preparing data for analysis
- Creating visual analytics
- Working with other visualizations and properties
- Controlling data selections
- Sharing insights, monitoring and finalizing the app

Task description

You are working as a business analyst for the human resource department in a company. The company has 7 offices across different countries.

There are 5 departments: marketing, legal, operations, finance, and HR.

Each department provides a list of its employees combined in an Excel file called **EmployeeHC.xlsx**. This file also contains information about the offices and the management team.

The human resource department wants to analyze the following metrics:

- Number of employees (including the management team) per office, country, department, age, gender, tenure, and job category
- Employee salaries based on department, gender, tenure (in years), job category, and education
- Office headcount and movements in the last year
- Office income and cost per location (country and city)
- Comparison between the American offices (Office1 and Office2) and English office (Office3 and Office7).

Create a Qlik Sense app for the company to analyze the office headcount and employee details. Answer the following business questions:

- What's the average salary in the company for each gender? And in the HR department? Compare the average vs the median salary. What is the salary distribution?
- Is there any correlation between the employee education or their tenure and the salary they earn?

- Which department is most likely to change its composition in the next 5 years? Discuss why and describe the changes you think will occur.
- Which country has the highest number of employees and what is the ratio of foreign employees in each country?
- Describe the headcount variation during the year for each office and the annual income. Also, display all employee details based on an office.

Process description

1. Review the data provided in the Excel file. Note the employee tables have the same structure but the field names may vary for each department.
2. Create a new Qlik Sense app with all the data. Make sure you transform the data, and combine and associate the tables based on the business needs.
3. Create as many sheets and visualizations as you need to cover all the business requirements, and analyze the data to answer the company questions. Advanced calculations might be required to answer some of these questions.
4. Use a story to describe your findings and tell the business the answers to these questions.

Apps and data files

- No apps are needed for this case. You will create a new app during the business case.
- There is only one data file called **EmployeeHC.xlsx**. You can download the file from the course page: **Student Files > Case3-AdvancedBusinessAnalysis**.

> **Qlik Sense Client-Managed**: If you are working with the Skytap environment, this file already exists under **C:\Modules_Create Visualization with Qlik Sense\CVQS_BusinessCases\Case3-AdvancedBusinessAnalysis**.

The file contains 7 tables:

- **Offices** (contains the list of offices, with all the details related):

Field	Description (type or format)
OfficeID	Unique office identifier for each office (integer) Links with the other employee tables.
City	Office city location (string)
InitialHC	Number of employees in the company at the start of the last year (integer)
NewHires	Employees who were hired by the company during the last year (integer)
TransfersIn	Employees transferred into the office from another office during the last year (integer)
TransfersOut	Employees transferred from the office into another office during the last year (integer)
Exits	Number of employees who left the company in the last year (integer)
OfficeIncome	Amount in dollars the office has generated in the last year (money, *$XXX*)
Taxes	Amount in dollars the office had to pay due to taxes in the last year (money, *$XXX*)

- **HR** (list of the employees from the HR department).
- **Finance** (list of the employees from the Finance department).
- **Marketing** (list of the employees from the Marketing department).
- **Operations** (list of the employees from the Operations department).
- **Legal** (list of the employees from the Legal department).
- **Management** (board of chief officers from all the departments, also known as the management team).

Other definitions from the employees tables are:

- **DOB** is the date of birth of each employee. You can calculate the age of the employee based on this field.
- **JoinDate** is the date the employee started in the company. You can calculate the tenure of the employee based on this field.
- **ForeignFlag** is the flag to identify if an employee is a foreign employee based on that office location (if flag is equal to 1).
- **Education** is the level of studies of the employee. They are classified into these groups: *None, Graduate, Bachelor, Master*, and *Doctor*.
- **JobTitle** contains the department and job category of the employee. The category can be *Assistant, Consultant, Director,* and *Officer*.
- The employee salary is calculated based on the following formula: **Basic Salary + Bonus - SalaryDeductions**.

Hints

- Use the Data manager to transform and associate the data.
  - Employees tables can be concatenated together.
  - Create new fields in the employee table: *EmpName, Age, Tenure, Salary, Department,* and *JobCategory*. You can use **Age()** and **ReloadTime()** functions to get the number of years between the dates to calculate the employee age and tenure.
  - Add the *Country* information from a separate table (using a manual entry table).
- Create several sheets and visualizations, including Scatter plot, Combo chart, Pie chart, Bar chart, Waterfall chart, Map, and Table.
- Use set analysis and variables if needed. They can be helpful when analyzing specific data or comparing specific data sets (for instance American vs. English offices).
- Some charts will need to use advanced calculations, like the **Aggr()** function that can dynamically group the data in a calculated dimension.
- Create a story with several slides to describe the findings based on the business requirements and questions.

Case 4: Visualizing data and storytelling

Duration: (~2 hour)

Summary:

- Create new visualizations in a Qlik Sense app based on the business requirements.
- Store bookmarks and snapshots then create a story to show your findings during analysis.

Topics:

- Creating visual analytics
- Working with other visualizations and properties
- Controlling data selections
- Sharing insights, monitoring and finalizing the app

Task description

As a business analyst, you are requested to maintain the business apps by including new visualizations and sheets.

These industry-specific apps, provided by the business, need to be updated following specific requirements.

The business would also like you to present your findings after analyzing the data with the newly updated visualizations.

Healthcare: Clinical Variation - Orthopaedics

App link: https://demos.qlik.com/qliksense/ClinicalVariationOrthopaedic

Required updates:

- Analyze the distribution of the following metrics:
  - ALOS per Facility and Clinician based on patients over 60 years old.
  - Potential LOS saving per Clinician and Complexity Level.
  - Age of the patients
- Display the *#Procedures* based on the following factors:
  - Gender composition
  - Facility and Urgency hierarchy
  - Comparison with *Complications Rate* and clinician
- Review the seasonality of the *#Procedures, #Diagnoses*, and *Avg complexity:*
  - Based on the weekday and month (using the *_date* field) dimensions.
  - Show the period of study as (min year - max year), example: "Period of study: 2012-2017"

Human Resources: Human Capital Management

App link: https://demos.qlik.com/qliksense/HumanCapitalManagement

Required updates:

- Analyze the average salary distribution based on these factors:
  - Regional salary by country
  - Distribution by country and job role
  - Distribution by tenure and department
  - Average salary for full-time vs. part-time employees
- Describe the headcount movements and balance including new hires and terminations:
  - Analyze the movements based on department, country, and recruitment channel
  - Calculate the percentage of new hires and terminations based on employment status (full time and part time)
  - Calculate the correlation between terminations and new hires based on the recruitment channel
- Display the distribution of employees based on age by department
- Compare the correlation between average salary and employee satisfaction per region: APAC, North America, and Europe
  - Display the aggregated metrics for #Employees, #Employees older than 40, Average Salary, Median Salary, and Employee satisfaction ratio by region, country, gender, and tenure group.

Sales: Executive Dashboard

App link: https://demos.qlik.com/qliksense/ExecutiveDashboard

Required updates:

- Analyze the regional CY Sales based on these factors:
  - CY Sales vs. LY sales per region
  - Top 5 products based on CY Sales per region
  - Distribution of CY Sales per region and product group
  - CY Sales trend based on the weekdays per each region
- Perform a basket analysis to identify:
  - Other product sub-group(s) related to specific product sub-group: we want to analyze what other products customers are buying when they bought a specific product sub-group.
  - The basket composition, displaying the percentage of the product subgroups based on the orders.
  - Which products, product groups, and product subgroups are most common in small baskets (orders that contain less than 5 products). Show the order numbers, products, and product categories (group and subgroup) of those products.
- Review the different segments and prepare a store analysis sheet with the following:
  - Comparison of sales and number of orders per each segment
  - Sales trend based on weekday per each segment

- Correlation between the average sales per day, the number of customers, and the number of products by segment

IT: Help Desk Management

App link: https://demos.qlik.com/qliksense/HelpdeskManagement

Required updates:

- Analyze the case duration by looking at:
  - Average case duration and distribution of the case duration overall.
  - Distribution of the average case duration by request time and department.
  - Average case duration based on high priority cases.
  - Average case duration for high priority cases by department or IT resource.
- Focus on the open cases:
  - Display the number of open cases by status and priority. Identify which status has a number of high priority open cases above the average (and compare with the median).
  - Review the seasonality of open cases based on weekday and month. Compare this for departments.
- For all open high priority cases opened in the current period (last year based on the data), display:
  - The total number of open high priority cases.
  - Comparison of that same number of cases based on department, status, and request type.
  - Correlation between those cases and the average case duration per IT resource (highlighting the correlation with the department).
  - The details of the number of cases across each case status, based on department, IT resource, and request type.

Process description

1. Choose one of the apps to perform this task. Open the app and familiarize yourself with it. You can work on several apps if time allows.
2. Read the task descriptions and, in a new sheet, create the visualizations you consider most effective.
3. Analyze the data. Take snapshots of the interesting facts you find in the app (and create bookmarks if needed) based on the business requirements.
4. Create a story to present to the company, including some slides describing your analysis and findings.

Apps and data files

- All the apps are available in the Qlik Sense demo site (demos.qlik.com/qliksense). In case you cannot access the demo site, the apps are also available for download under Student Files.

  Qlik Sense Client-Managed: If you are working with the Skytap environment, the app or apps will be available under the **Everyone** stream.

- No data files are needed for this business case.

Hints

- Review all the sheets, visualizations, stories, and bookmarks (if available) in the app.
- Create a new sheet and include the most effective visualizations to represent the business needs based on the requested updates.
- Discover some interesting facts and information in the app you can explain or present to the business using storytelling.
  - Analyze the visualizations, making selections if needed, and take snapshots to include in the story.
  - If you want to point to a sheet with a starting selection point, create a bookmark and include it as a hyperlink in your story.
  - A complete story should include snapshots, effects, shapes, explanatory text and titles, links to bookmarks in the app, and other sheets in the app to analyze more details.

Case 5: Director sheet (Vintage Movies)

Duration: (~1 hour)

Summary:

- Create a sheet with advanced calculations in a Qlik Sense app.
- Use variables to reuse parts of expressions and measures.
- Apply set analysis to visualizations to focus the analysis on a specific dataset.

Topics:

- Creating visual analytics
- Working with other visualizations and properties
- Controlling data selections

Task description

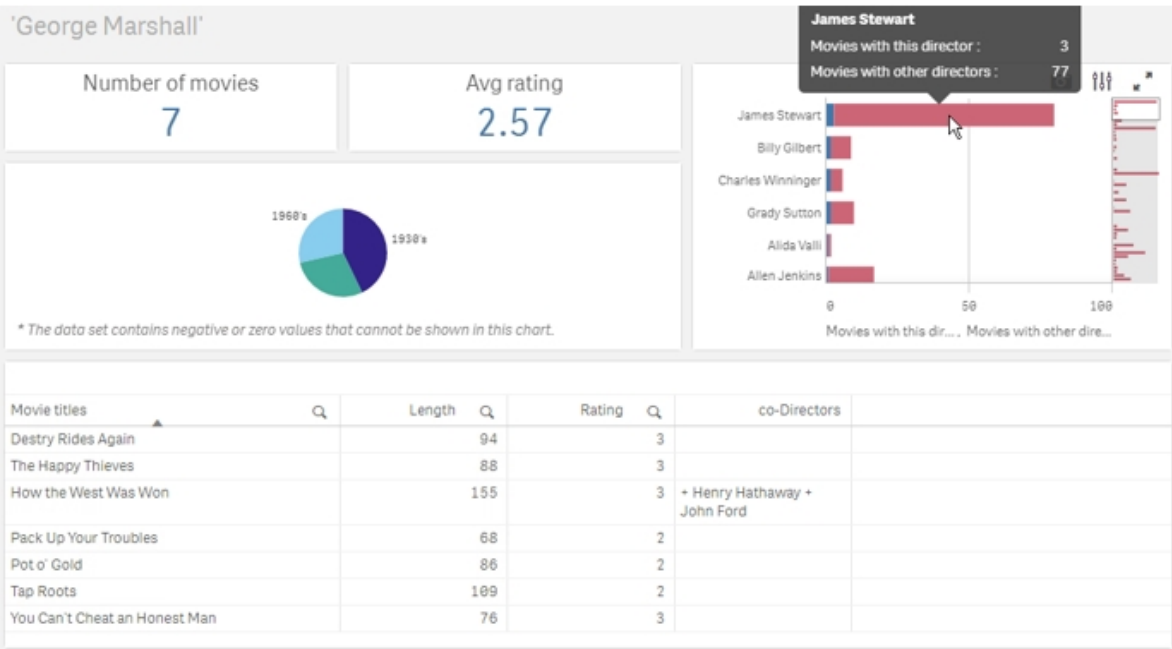
We would like to create a new sheet in the **Vintage movies** app dedicated to one of our favorite directors. Feel free to choose one of the directors in the app (i.e., *Steven Spielberg*).

The new sheet should include an analysis of this director using the following data elements and suggested visualizations:

- Number of movies and average rating (as a KPI)
- Movies with this director vs. movies with other directors by actor (as a Bar chart). Only display actors working with the director
- Percentage of movies across the decades (as a Pie chart)
- Movie titles, length, rating, and the list of other directors who co-directed the movie if applicable (in a Table)
- The sheet name should display the director's name

The director can be changed at any point, so you can decide to present the same sheet with a new director's information.

Example - Change the director to ‘George Marshall’ and compare your results with this example:



Consider using variables to define parts of the expressions as different visualizations can reuse these elements.

Process description

1. Open the Vintage Movies app and create a new sheet. For each visualization, you need to list the dimensions and measures.
2. Discuss the areas in the required measures and dimensions you can reuse or store in variables.
3. Create the required variables for re-usability in your calculations. Take into consideration the requirements of changing the sheet to a new director at any point.
4. Create and configure the new visualizations with the dimensions and measures (some of them will include set analysis and variables).
5. Make selections to analyze and test your expressions. Then change the variables to see how you can change the director.

Apps and data files

- Download the file of the **Vintage movies** app from the course page: **Student Files > Case5-DirectorSheet**.

  Qlik Sense Client-Managed: If you are working with the Skytap environment, this file already exists under **C:\Modules_Create Visualization with Qlik Sense\CVQS_BusinessCases\Case5-Director sheet (Vintage Movies)**.

Hints

- You can use the existing master measures in the app, and then, unlink them and apply set analysis.
- Use variables in your set analysis for re-usability. Use two variables: **vDirector** and **vSetAnalysisDirector**:
  - **vDirector** stores the director name in single quotes as plain text. You can change the director by just changing the value in this variable.
  - **vSetAnalysisDirector** contains the set modifier content: *Director={$(vDirector)}*.
- **Aggr()** and **Only()** are used to apply set analysis in dimensions. For instance, to make sure the bar chart only displays the actors working with the director and not all the actors in the app, you might need to use a calculated dimension with **Aggr()** and set analysis.
- Use the **concat()** function to display the list of co-directors and the **replace()** function to avoid displaying the main director name.
- For the bar chart calculation, you might need to use a set operator (i.e., *1-$*).

Case ABC: Summary

Duration: (~6 hours)

Summary:

- Introduce the activity scenario for the ABC Apparel company.
- Review and understand their data and business requirements.
- Create a Qlik Sense app file, then load, transform, and associate data from an Excel file.
- Build a dashboard sheet as the first sheet in the app file, then create additional sheets that will answer the specific needs of the users of the Qlik Sense app.

Topics: You will apply what you've learned throughout this course in this business case:

- Preparing data for analysis
- Creating visual analytics
- Working with other visualizations and properties
- Controlling data selections
- Sharing insights, monitoring and finalizing the app

Task description

The ABC Apparel Inc. (ABC company) just invested in a new business intelligence tool, **Qlik Sense**, and they have asked you to create their first Qlik Sense app, **ABC Sales Dashboard**.

The company has prepared a business plan for you to understand the company background, their data, as well as their needs and business requirements.

The business plan

The company background

ABC Apparel Inc. (ABC company) is an international clothing company based in Lund, Sweden.

- ABC is a wholesaler of fine clothing, buying from a variety of manufacturers and selling to retail store outlets.
- They have five sales offices in four countries: Lund (Sweden), Hong Kong (China), Nice (France), New York and Seattle (USA).
- The corporate headquarters in Lund has these departments: Marketing, Sales, Finance, IT, Accounting, Purchasing, and HR.
- The company has about 1,200 customers in 21 countries. Customers are a mix of small boutiques and large retail department stores.

ABC is enjoying rapid growth in their business. The combination of good products, excellent service, and competitive prices have provided them a solid value proposition that customers eagerly embrace. As with any growing company, however, there are many challenges. Company leadership is focused on several important efforts to position them for a future of growth and prosperity.

The company is looking for a Dashboard that provides analysis for sales.

One of those efforts is the implementation of state-of-the-art business systems. The IT Director has undertaken this challenge and has been replacing and improving the IT infrastructure over the last three quarters. One part of this effort includes addressing some weaknesses in their business reporting and analytics:

- There is no single integrated BI system. Each business system provides some basic reporting. They are not integrated with one another in any meaningful way. This makes cross-system reporting difficult or impossible.
- Standard reports from existing systems are too limited. They only offer rudimentary reporting and no analysis. There are only limited possibilities to drill down and ask new questions due to pre-designed static reports and limited dimensionality.
- Reporting is done on an ad-hoc basis. Each person, department, or group creates and maintains their own set of reports, resulting in much duplicated effort and report latency. Inconsistency of KPIs and measures leads to confusion.
- Static reports are not up-to-date. Business managers often work from old reports.
- There is a low business user adoption rate. Business users do not have a reliable, convenient way to get their reports, so they don't use them or use old ones.
- Maintenance of standard reports, systems, and data are out of IT's direct control.

The company is looking for an easy-to-use, robust, flexible, and powerful platform that serves as the foundation for self-service data discovery.

- They need to analyze their data from a summary to detail level.
- They need a system that can integrate their various data sources into "one version of the truth."

ABC has selected Qlik Sense as the foundation for their reporting and analysis needs.

The Sales application

The Sales app is the first app ABC is going to launch using Qlik Sense:

- It is targeted to the CEO and CMO together with their direct reports.
- The Sales Manager was involved in planning and, being a Qlik Sense champion, wants to extend the initial app to fit the sales team's needs and demands.

A single app is required for these three users. Each of them would have a different user experience and different business needs:

The CEO (Chief Executive Officer)

The CEO is responsible for:

- Aligning the company's strategic vision internally and externally.
- Facilitating business outside of the company, while guiding employees and other executive officers towards global objectives.

This user:

- Works with the app on a weekly basis, and needs a prompt, thorough, and accurate overview of the company's key performance indicators.
- Might access the app using a tablet or a mobile device.
- Needs to provide information to keep the board appropriately informed of the organization's financial position.

- Is not going to modify the app.
- Needs an immediate overview without having to perform complex searches.

Measures and dimensions for this user are:

- Trend of total sales, and comparison between CYTD (Current Year to Date) and LYTD (Last Year to Date) sales
- Sales share by Region, Division, and Product category
- Number of orders CYTD
- Margin percent by product group and by region

This user would need to answer the following business questions:

- What is the year-on-year comparison for monthly sales, CYTD vs LYTD?
- What does the sales trend look like?
- How do the CYTD sales vary by division or region? Are there regions that are performing well in sales that can offer help (with regard to top products, and so on) to regions that are not performing that well?
- Has the order volume improved since the previous year/month?
- What is the margin% we have per each region and division?

The CMO (Chief Marketing Officer)

- Is responsible for activities in the organization that have to do with creating, communicating, and delivering offerings that have value for their customers.
- Facilitates growth and increase sales by developing a comprehensive plan that will promote brand recognition and help the organization gain a competitive advantage.
- Uses the app on a daily basis on a laptop for high level analytics but also to analyze product and sales performance on a more detailed level.
- Has experience with Qlik Sense and knows how to modify existing charts and create new sheet objects based on the underlying data model.

Measures and dimensions for this user are:

- CYTD Customer sales versus LYTD
- Product sales and order volume
- Trends on Sales, COGS, Margin%, Number of orders
- Number of orders per product category
- Last orders placed

This user would need to answer the following business questions:

- The Marketing department would like to run a campaign highlighting top-selling products in the regions. Therefore, it needs to determine which products are top sellers in each region. Does this differ by region?
- Marketing would like to reach out to customers who have not made a purchase in the past six months. They need to determine who these customers are. They also need to see if these customers are platinum, gold, or silver customers based on ABC's classification.
- What is the number of orders each customer has made and what is the value of these orders? Is there a correlation between these two?

- Are there many customers who have less sales this current year to date than last year to date? The CMO needs to view the customer details of these customers. Did these customers receive discounts with their orders?
- In past years, Swimwear showed poor sales. This year, a new collection is launched and Marketing has to identify the major target group for a marketing campaign.

The Sales Manager

- Is responsible for leading and guiding the global Sales teams according to the company's goals.
- Uses the app on a daily basis and needs to create and share reports as a basis for one-to-one meetings with sales people.
- Works primarily remotely and accesses the app from a laptop.

Measures and dimensions for this user are:

- Regional sales based on customer country and city
- Top and bottom products and customers – based on sales
- Region and division performance over time

This user would need to answer the following business questions:

- What are the top five and the bottom five selling products?
- Who are the top customers for a selected period?
- Where are these customers based and what is their order volume and average deal size?
- How is the discount related to product and customer?

The business requirements

ABC Apparel Inc. is interested in analyzing their data over time.

- Time needs to be displayed based upon the Order Date field in the dataset.
- Analysis must be possible based upon Year, Month, Year Month, and Quarter values for the entire dataset.

Graphical components

The marketing department has provided you with a graphical style guide document and created a script file with definitions of the company's corporate colors. You can find sheet backgrounds, buttons, and logos in the **Images** folder.

Main dimensions

In order to filter the data, these fields are defined as the main dimensions:

- Customer
- Division
- Region*
- Country
- City
- Sales Person - First Name and Last Name of the employee

- Product Category - Category name of the product
- Product
- Time dimensions (Year, Month, Quarter, Year Month) based on Order Date.

*Region is based on divisions, where ABC corporation has defined that *South America* and *North America* are grouped as Americas.

Customers are also classified in the following groups based on their average deal size (dynamically based on selections):

- Platinum - customers with more than 5.5k avg deal size.
- Gold - customers with avg deal size between 4k and 5.5k.
- Silver - customers with less than 4k avg deal size.

Key measures

ABC has defined the following Key Performance Indicators (KPIs) and measures for the analysis application.

Providing measurement visibility indicates compliance with business goals through the various ways they are presented in the analysis tool.

Ensure that these measures have the ability to be trended over time.

Master measure name	Expression
Sales	`Sum(Sales)`
COGS	`Sum(UnitCost * Quantity)`
Discount	`Num(Avg(Discount),'0.00%')`
Margin	`Sum(Sales - (UnitCost * Quantity))`

Margin%	`Num(`
	`Sum(Sales - (UnitCost * Quantity)) / Sum(Sales)`
	`,'0.00%')`

Num Orders	`Count(Distinct OrderID)`
Avg Deal Size	`Sum(Sales)/Count(Distinct OrderID)`

Include the current year to date measures for Sales and number of orders.

Compare as well LYTD Sales with CYTD Sales using the following calculation: CYTD - LYTD Sales

User interface and enhancements

The Dashboard app should include the following sheets or sections:

- Dashboard sheet - main sheet with the high level details of the main KPIs for the users
- Product sheets - analyze sales and order volume by product and product category
- Customer sheets - analyze sales and order volume by customer
- Sales sheets - analyze sales, margin and the sales team
- Regional sales and orders sheets

- Comparative analysis sheet - using different selection subsets, compare sales and order volume

Other app enhancements include the following:

- Apply custom coloring based on product category.
- Include a story in the app about EMEA division. This will be a story to present to the marketing team.
- Create bookmarks for the end users to select the current year and the EMEA or Americas region.

Process description

1. Review the business plan provided by the ABC company.
2. Discuss the steps required to achieve the business needs in the plan.
3. Create a new app and load, transform, and associate the data from the Excel file provided.
4. Create a front-end user interface using sheets and visualizations. Choose the right visualization to represent the business measures and requirements.
5. When needed, apply calculated conditions, advanced expressions, set analysis, and other enhancements such as colors, presentation, functionality, etc.
6. Make selections to analyze and test your solution.

Apps and data files

- Use the Excel File: **SalesData_MultiTabs.xlsx**. This file is available in the course data files.

  Qlik Sense Client-Managed: If you are working with the Skytap environment, this file already exists under **Qlik Sense Data**.

Hints

- Review the business plan and understand the end user's and the company needs.
- Discuss the business requirements before you start the app development.
- Follow your instructor's guidance and visit the Qlik Help site when needed.
- Apply the visualization concepts when creating the dashboard to choose the right visualization.
- Use the activities in the course for extra guidance.

Case ABC 1: Product stock (ABC Sales)

Duration: (~1 hour)

Summary:

- Add additional data from Excel to a Qlik Sense app.
- Transform and associate with the rest of the tables. Use the Unpivot feature.
- Visualize the new data based on business requirements.

Topics:

- Preparing data for analysis
- Creating visual analytics

Task description

We would like to include the Stock by product size information provided by the warehouse in the **ABC Sales Dashboard** app. This information is provided in an Excel file with the following format, where:

- The first column represents the product (product name). The warehouse uses a different product identifier than the ID used in the **ABC Sales Dashboard** app. In this case, the business uses the product name as the unique ID per product to associate with the sales data.
- The columns *XS, S, M, L, XL* (highlighted in green) represent the clothing sizes and show the corresponding number of product units available in the warehouse for each size and product.
- The columns *0-2, 3-5, 6-8, 9-11, 12-14* (highlighted in gray) represent the shoe sizes and show the corresponding number of product units available in the warehouse for each size and product.
- *Total Stock* is the sum of all the stock units for all sizes per product.
- *Description* shows the product category details of each product.

Product-Size	XS	S	M	L	XL	0-2	3-5	6-8	9-11	12-14	Total Stock	Description
AAA Running Shoe						53	59	2	90	33	237	Sports clothing
Aino Shoes						57	21	0	0	0	78	0-3 Years
Atles Lussekofta	82	67	74	97	24						344	Fashion for Men
Baby Dark Lounge Suit	14	27	65	23	0						129	0-3 Years
Basket Shoes						70	64	66	23	41	264	Sports clothing
Basket Vest	25	88	81	42	43						279	Sports clothing
Baywatch Swimsuit	24	11	80	70	100						285	Beach clothing
Bike Helmet	79	20	89	86	22						296	Sports clothing
Bow tie	0	0	66	0	0						66	Fashion for Men

The new sheet should include an analysis of the warehouse product stock data together with the product sales. Some of the key measures and factors the business would like to analyze are the following:

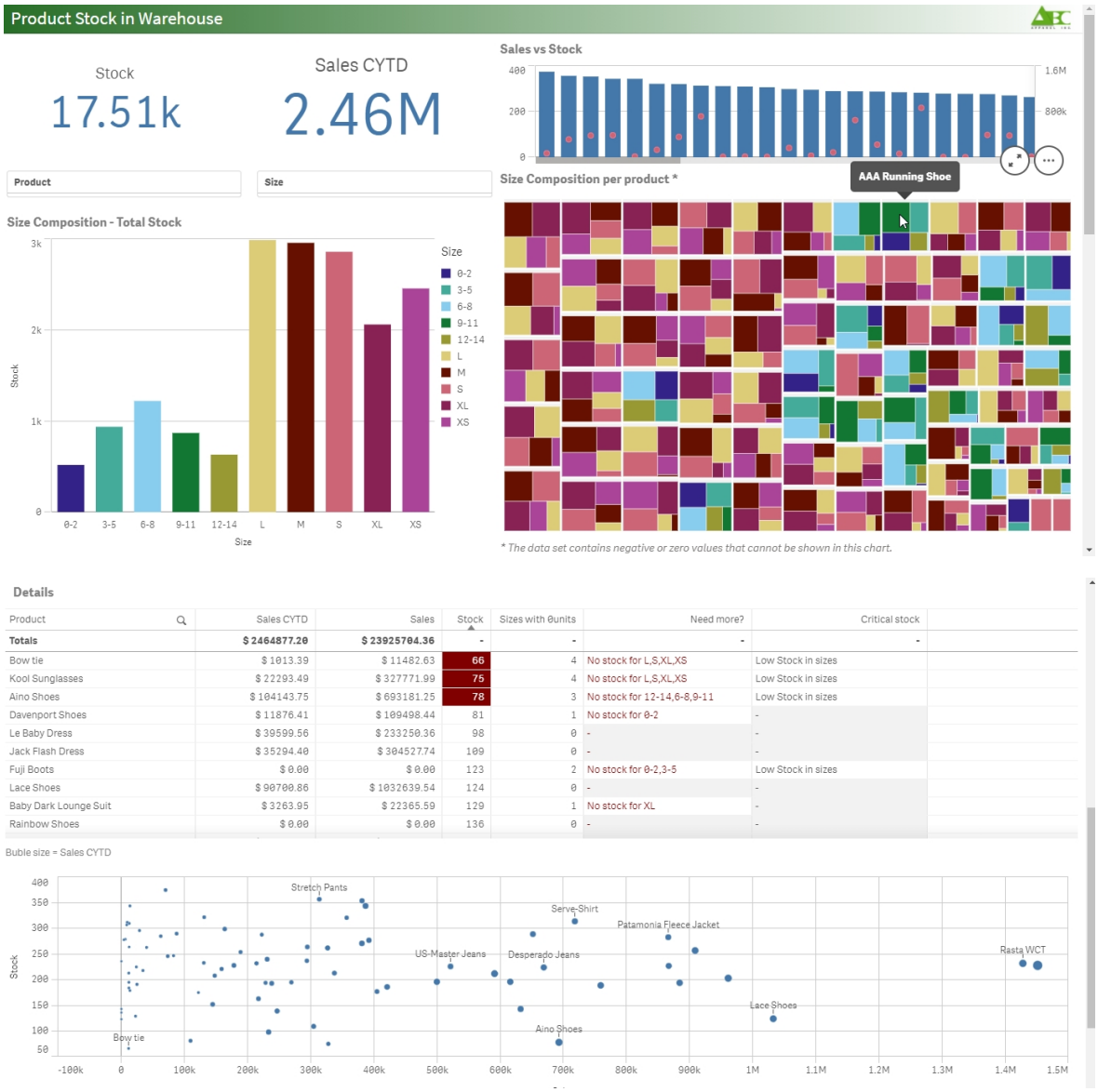
- Review the product sales, sales CYTD, and product total stock comparison.
- Monitor the total stock and the size volume. Identify these values per product.
- Flag the products that need more stock, where the units of the product sizes run out (0 units). Also, identify in how many sizes there is no stock.

- Identify Critical stock items:
  - Products with a high level of units (total stock over 300 units) and a very low level of sales (lower than 300k).
  - Products with no units in at least two sizes or the total stock is lower than 80 units.
- Perform a size search - select one size and see all the products with available units in the selected size.

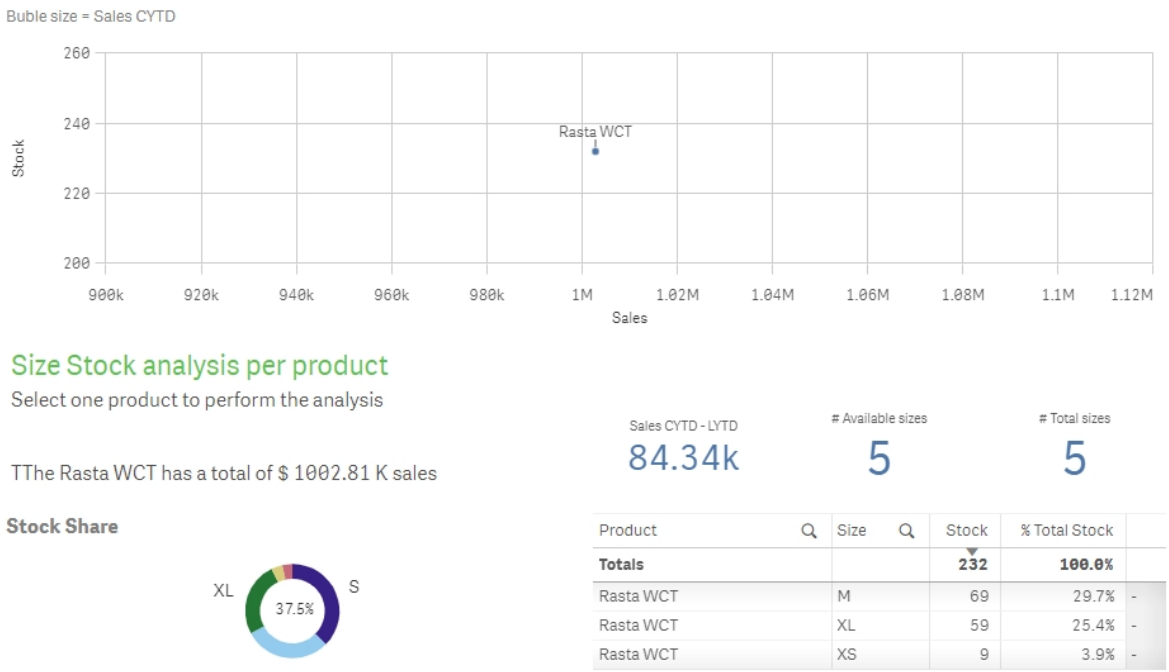
The business would like to analyze, for each product, the stock details when only one product is selected as follows:

- Percentage of stock sizes based on the total stock of a product
- Number of available sizes of the product
- Flag sizes that need more units (where there are less than 5 units in a size)

Example - The following example shows a dashboard the business would like to have to monitor the warehouse stock:



When selecting a product, you can also see the product details and the size analysis based on stock.



You can create you own version of the dashboard, as long as it covers the business requirements.

Process description

1. Open the Excel file and the **ABC Sales Dashboard** app to review the data model and the new data.
2. Discuss the steps you will need to follow to load, transform, and associate the new data. Then include the new data in the app.
3. Create any flags needed in the new table to monitor the business needs using the transformation cards.
4. Create and configure the new visualizations in a new sheet in the app called **Product Stock in Warehouse**.
5. Make selections to analyze and test your expressions. Then discuss your solution with the class.

Apps and data files

- Use the **ABC Sales Dashboard** app and the **ProductSizeStock** Excel file. They are available in the course data files.

> **Qlik Sense Client-Managed**: If you are working with the Skytap environment, this file already exists under **Qlik Sense Data**.

Hints

- You can load the Excel file, unpivot the size columns, and create new flags to identify what sizes have 0 units.
- Use the **Unpivot** feature. Keep the **Product-Size** columns and **Description** and create two new columns:
  - **Size** contains the available sizes.
  - **Stock** contains the number of units.
- Create a calculated field, *Flag_LowStock*, to identify if a *Size* has 0 stock. This field will contain a 1 or 0 flag value that can be useful when creating the expressions in the front end.
- Create visualizations with calculated expressions to cover the business needs.
- Review the solution app for this business case if needed.

Case ABC 2: Customer thresholds (ABC Sales)

Duration: (~20 min)

Summary:

- Modify existing visualizations in an app.
- Create reference lines (thresholds) in a chart with advanced expressions.
- Use variables to make dynamic expressions.

Topics:

- Creating visual analytics
- Working with other visualizations and properties
- Preparing data for analysis

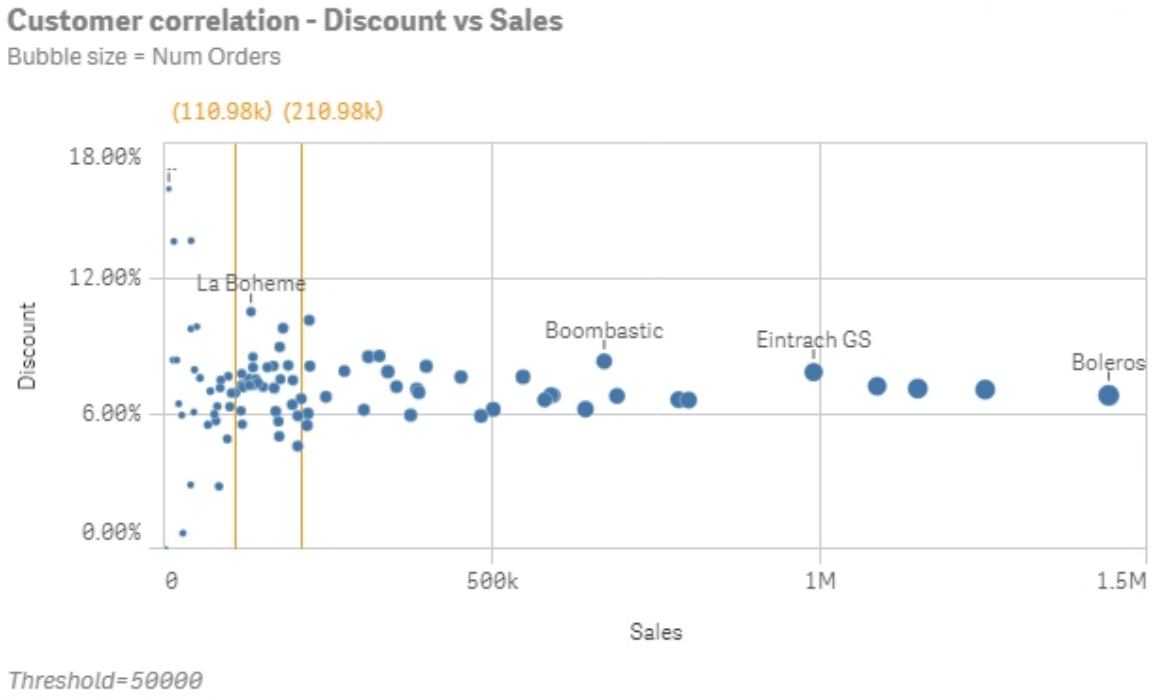
Task description

We would like to enhance the UI in the **ABC Sales Dashboard** app.

The business has requested the following enhancements in the Customer sales sheet:

- Include dynamic reference lines in the existing scatter plot as follows:
  - Line 1 should show the median of the sales per customer plus a fixed amount as positive threshold*
  - Line 2 should show the median of the sales per customer minus a fixed amount as negative threshold*
  - *The *threshold* is a value defined by the ABC company that will change over time. Create and apply a variable to set this amount initially to be 50k.
- Use a single variable, called *vThreshold*, with a fixed value to store the defined threshold by the business. Users will be able to change this value.
- Include a footnote in the chart to clarify the *threshold* value.
- Allow users to change the *vThreshold*, without having to edit the variable, using these values: 25k, 50k, 100k, and 150k.

Example



As many of these visualizations will have parts of the same analysis, use variables to reuse parts of the expressions that can be applied to all.

Process description

1. Open the ABC Sales Dashboard app and locate the Customer sales sheet.
2. Discuss the steps required to achieve the business needs.
3. Create a *vThreshold* variable for re-usability in your calculations.
4. Create and configure the new expressions for the reference lines in the existing scatter plot.
5. Include new objects in the sheet to allow the users to change the variable values. Take into consideration the requirements of changing the threshold at any point with the specified values.
6. Make selections to analyze and test your solution.

Apps and data files

- Use the **ABC Sales Dashboard** app. This app is available in the course data files (under apps).

  Qlik Sense Client-Managed: If you are working with the Skytap environment, the app or apps will be available in the Hub.

Hints

- Reference lines can be added to scatter plots from the properties panel.
- In order to calculate the median of the sales per customer, you can use the **Aggr()** function to group by customer and then apply the **Median()** function.
- The threshold fixed value can be defined using a variable, as it is being used in two reference lines. The user can change this threshold later on without having to edit the expressions in the reference lines.
- For the bonus point, you can discuss different options:
  - Review the Qlik Dashboard bundle in Custom objects section (if available), for variable input, or the Charts section for buttons with actions.
  - Or use a manual entry table with the defined values, list the values in a filter pane, and use the single value selected to always have one of them selected.

Case ABC 3: Geo analysis (ABC Sales)

Duration: (~20 min)

Summary:

- Create new sheets and visualizations in a Qlik Sense app.
- Configure different map layers to represent regional measures.

Topics:

- Creating visual analytics
- Working with other visualizations and properties

Task description

The Marketing team has requested new visualizations in the **ABC Sales Dashboard** app.

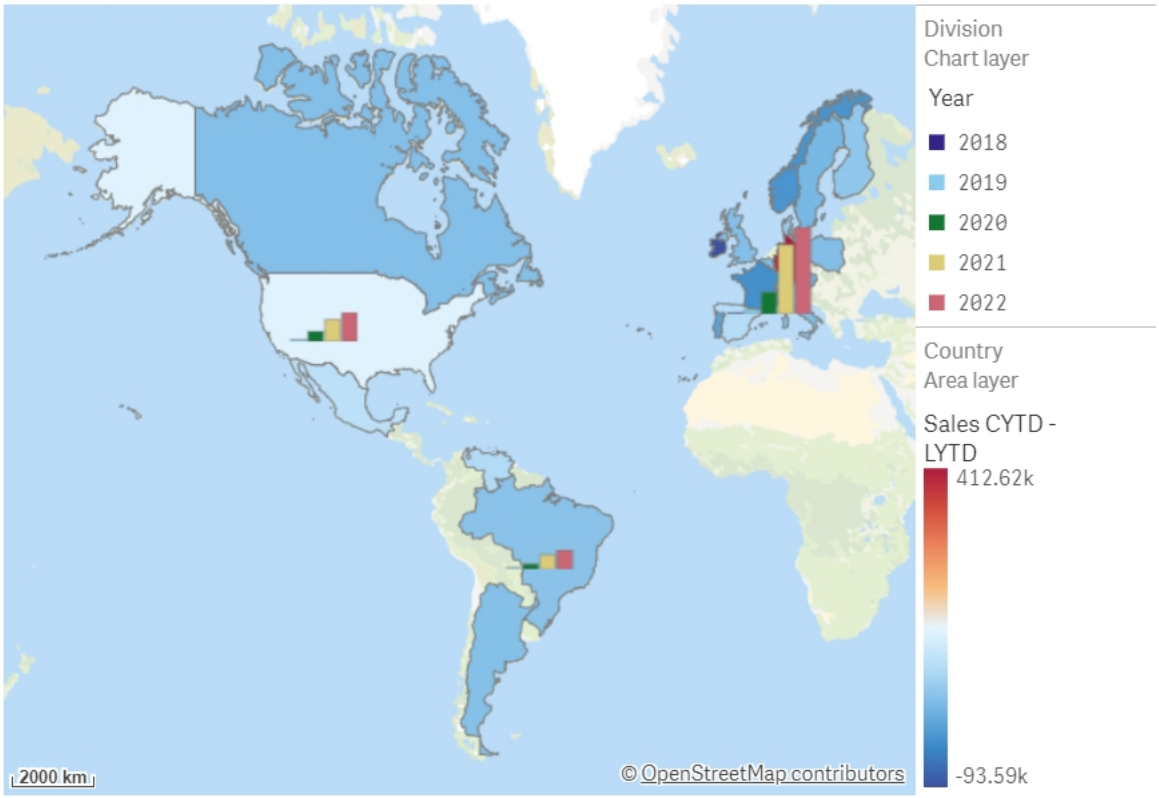
The team would like to perform a geographical analysis based on the customer division, country, and city.

The **Regional Sales YTD comparison** map should identify the following information on the map:

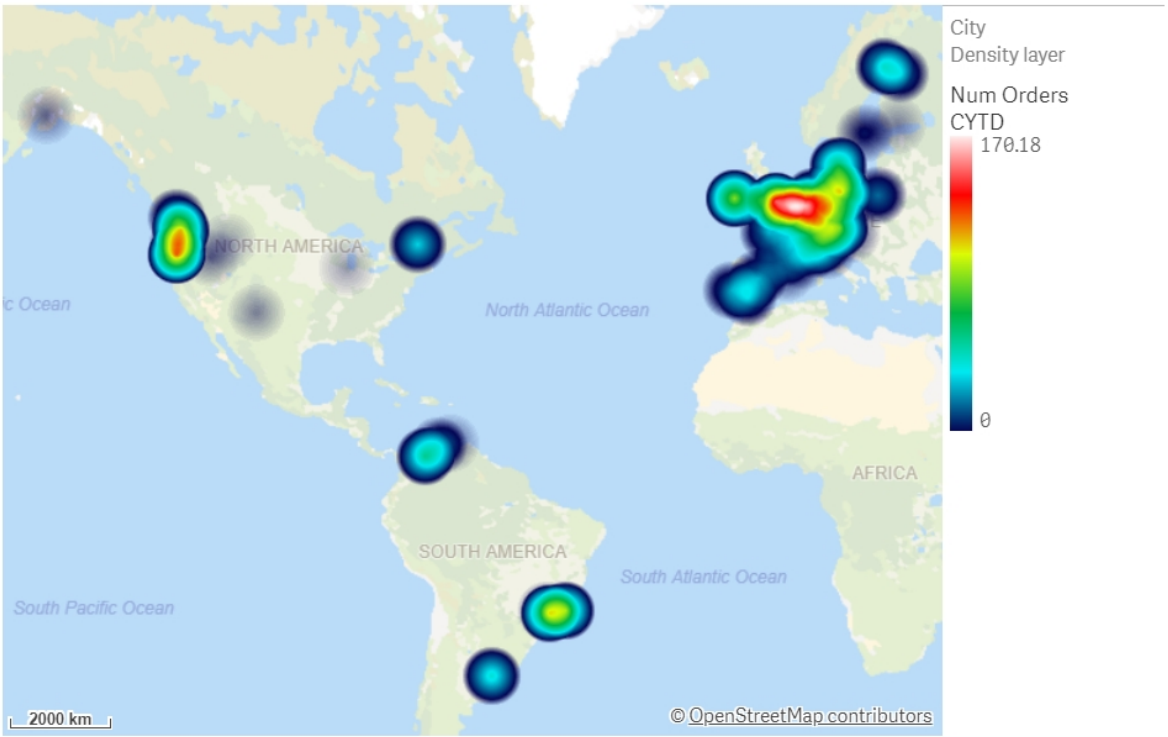
- Sales YTD by Year and Division
- Sales CYTD - LYTD by Country
- Order volume CYTD by City (representing density)

Examples

Sales Current Year vs Last Year to date



Order Volume CYTD



Maps can include different layers, use the ones that you consider more appropriate based on the business needs.

Process description

1. Open the ABC Sales Dashboard app and create new sheets if needed.
2. Discuss the steps required to create the new visualizations.
3. Create two maps with the required layers, measures, and dimensions.
4. Configure the Map properties.
5. Make selections to analyze and test your solution.

Apps and data files

- Use the **ABC Sales Dashboard** app. This app is available in the course data files (under apps).

  Qlik Sense Client-Managed: If you are working with the Skytap environment, the app or apps will be available in the Hub.

Hints

- When creating the maps, add the following layers:
  - Chart for Sales CYTD by Division
  - Area for the Sales CYTD - LYTD by Country
  - Density for the Num Orders CYTD by City
- You can combine two or more layers in a map, but keep the maps clean so the user can analyze the data. If there are too many layers in one map, you can use more than one map object.
- The *Country* and *City* dimensions are already geofields. The *Division* dimension will need to use expressions to apply the geolocation with latitude and longitude. Use the **Avg()** function to calculate the average latitude and longitude.

Case ABC 4: Year on year analysis (ABC Sales)

Duration: (~20 min)

Summary:

- Create visualizations and use color expressions in a Qlik Sense app.
- Use set analysis to define the year on year analysis in the expressions and product comparisons.
- Use variables to define dynamic dimensions.

Topics:

- Creating visual analytics
- Working with other visualizations and properties
- Controlling data selections

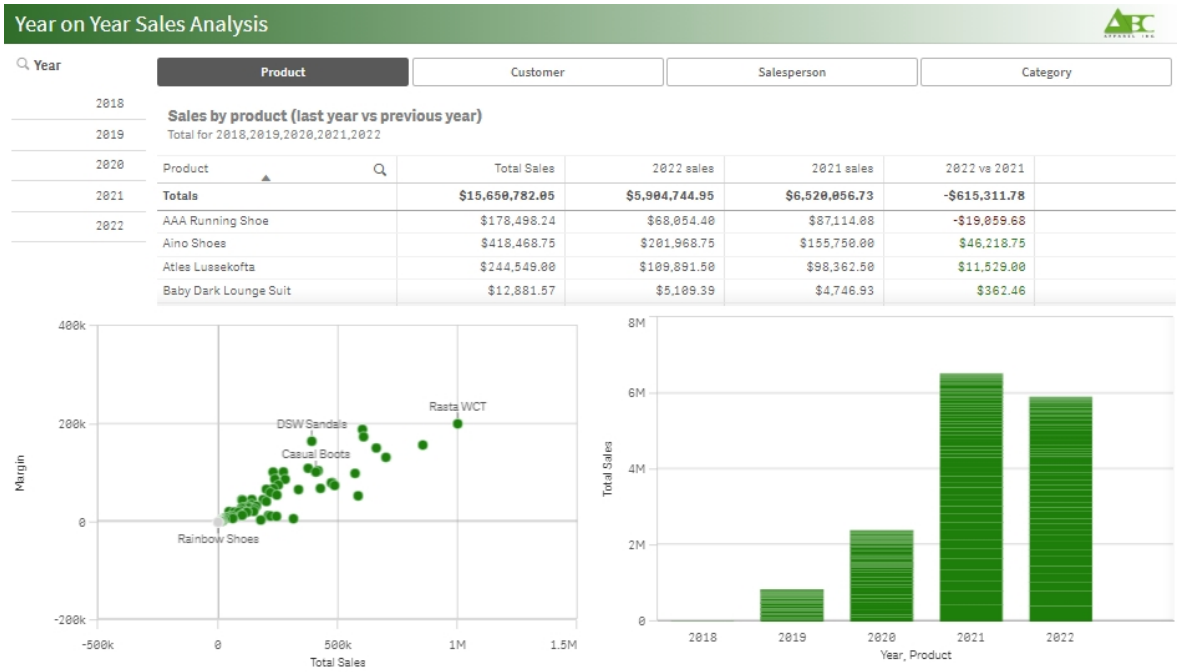
Task description

The CEO of the company has been using the ABC Sales Dashboard for a while now and he has requested a second dashboard to allow a year on year analysis with some specific requirements:

- Include a table to show the following measures or KPIs:
  - Sales based on user selections - this can include several years.
  - Sales for the last year to date based on user selections.
  - Sales for the previous year based on user selections.
  - Comparison between sales last available year and previous year. Include text with custom coloring for the comparison.
- Include correlation between total sales and total margin based on a dimension* (use color to highlight the current selection).
- Include a total sales year on year comparison based on a secondary dimension* (use color to highlight the current selection).

*All the visualizations are based on a dimension: *Product* (by default) that can be changed by the business user at any time for the following dimensions: *Customer, CategoryName*, and *Salesperson* (*EmpName*)

Example



The suggested example by the CEO uses a row of buttons to change the dimension in the charts and table. You can apply this solution or any other similar solution to your final dashboard.

Process description

1. Review the CEO requirements and discuss with the class how to achieve the desired behavior.
2. Discuss and think about the set analysis required for the expressions.
3. Open the ABC Sales Dashboard app, create a new sheet and a new variable.
4. Create suitable visualizations and use advanced expressions to apply set analysis and color by expressions.
5. Apply a solution to allow the user to change the dimension in the visualizations using the variable.

Apps and data files

- Use the **ABC Sales Dashboard** app. This app is available in the course data files (under apps).

> **Qlik Sense Client-Managed**: If you are working with the Skytap environment, the app or apps will be available in the Hub.

Hints

- Use set analysis in expressions, applying set identifiers and modifiers, like in the following examples:

- All data: *{1}* - selections will not affect the calculation
- Current selections: *{$}* - this identifier can be omitted
- Current selections but displaying all products: *{$<Product=>}*
- Current selections for a specific year (2021): *{$<[OrderDate.autoCalendar.Year]= {2021}>}*
- Current selection for the latest available year: *{$< [OrderDate.autoCalendar.Year]= {$(=Max(Year(OrderDate)))}>}*

- Use color by expression in the visualizations to color only the current selections: *if(sum (Sales)>0, green())*
- Use a variable to define the dimension in the table and charts. Then include a solution that allows users to change the variable dynamically. You can use the **Variable Input** extension or if not available, a filter pane with a data island.